Chapter 4

Application of tipping point techniques to tropical cyclones

In this chapter, we apply the techniques described ~~in~~ previous ly ~~chapters~~ to the real geophysical problem of providing an early warning signal for the approach of a moving tropical cyclone [1]. Typically, a steep drop in sea-level pressure is observed at the arrival of a cyclone, as shown in figure 4.1. This sudden change in the state of the system is not bifurcational in the sense that there are multiple stable system states, but it is nonetheless a tipping event, and the point at which it happens may be referred to as a tipping point. Similarly, there is a steep rise in wind speed at the same time. The hypothesis is that it is possible to detect or predict this observed tipping point using the same methods previously used to analyse examples of tipping points known to be the result of a switching between stable states or bifurcations.

Variables

Large tropical cyclones extend hundreds of kilometres from the centre and a location on the future trajectory of such a storm experiences high winds several hours or days before the arrival of 200km/h winds. We expect, therefore, that the study of the weather at a fixed location will provide an early warning signal for the arrival of a tropical cyclone. The rise in wind speed itself acts as such a signal, but we interested in the possibility that the tipping point techniques described in chapters 2 and 3 will provide an early warning signal when applied in this context.

In section 4.1 we give a description of the data used throughout this chapter, which has been obtained from the NOAA HURDAT2 database and the Met Office HasISD database. [REF] In section 4.2 we apply the ACF1, DFA and PS indicators (see chapter 2) to the sea-level pressure time series data from locations on the paths of fourteen tropical cyclones. In section 4.3 the

[1] A tropical cyclone is referred to as a 'hurricane' when it occurs in the Atlantic Ocean and north-eastern Pacific Ocean and a 'typhoon' when in the north-western Pacific Ocean. In the south Pacific or Indian Ocean, comparable storms are referred to simply as 'tropical cyclones' or 'cyclones'. These naming conventions are not always abided by (e.g. Australian "hurricanes") and do not denote any differences in the physical properties of the storms other than geographic location.

same fourteen points are used but both the sea-level pressure and the wind speed variables are considered and the techniques described in chapter 3 are applied to the two variables.

In section 4.4 we use sea-level pressure time series from several locations in a region through which a tropical cyclone passes. We apply the techniques introduced in section 3.6 to visualise the rise of a tipping point indicator over the area as the tropical cyclone approaches. This idea is further developed in section 4.5 where a model of the approaching cyclone is presented ~~in order to provide an additional test of the technique.~~

4.1 Description of data

4.1.1 The HURDAT2 database

During this study it was necessary to have track data of several Atlantic hurricanes, this was obtained from the NOAA national hurricane center's HURDAT2 database[2] [Landsea and Franklin, 2013]. This dataset has a comma-delimited, text format with six-hourly information on the location, maximum winds, central pressure, and size of all known tropical cyclones and subtropical cyclones. We have only used the location data (given as lat-lon coordinates) and have not performed any additional processing of this data.

4.1.2 The HadISD database

For the purposes of our study we use sea-level pressure (SLP) and wind speed data from weather stations. These data are obtained from the HadISD 2017 data set[3] [Dunn et al., 2012, 2014, 2016; Smith et al., 2011] maintained by, and available from, the UK Met Office. The sea-level pressure and wind speed data provided by HadISD are raw data given at one-hour intervals, the Met Office applies a filter so that weather stations with ~~providing~~ very sparse data are disregarded, no other processing is applied ~~used~~ [Dunn et al., 2012].

The weather stations used in this study are selected for proximity to the landfall locations of tropical cyclones, and data are considered in a fifteen-day (360 hour) period before the time when the cyclone passes closest to the station. However, some of the weather stations provide very sparse data even after the Met Office's filtering, this is common close to the time when a tropical cyclone passes in proximity of ~~close to~~ a station (presumably due to infrastructural damage). If fewer than 80% of the time points have associated data, the time series is disregarded. Otherwise the data are ~~is~~ interpolated linearly onto the entire hourly time series. We notice~~d~~ that weather stations on the United States mainland are more numerous than in other regions affected by tropical

W

[2] see https://www.nhc.noaa.gov/data/#hurdat

[3] Access to the data set is available at www.metoffice.gov.uk/hadobs/hadisd/v202_2017p

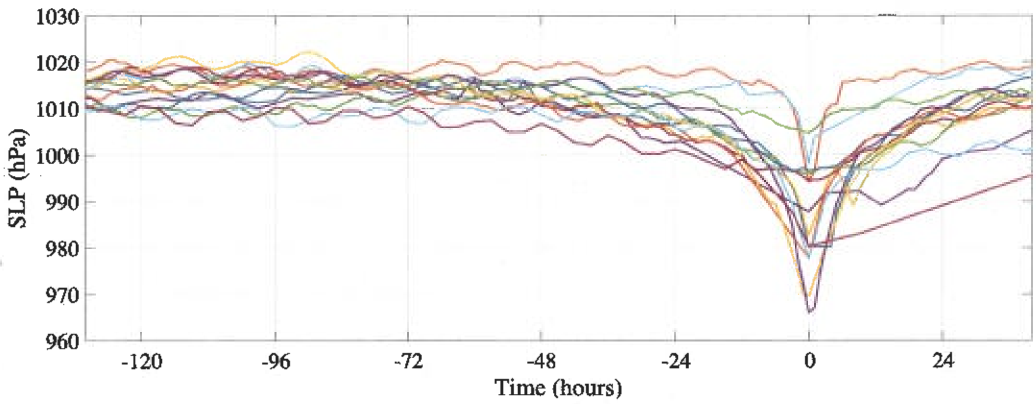


Fig. 4.1 Sea-level pressure ~~(SLP)~~ time series for the fourteen tropical cyclones selected in table 4.1, the minimum point in the data is presumed to be the point closest to the time at which the cyclone was closest to the station and is set as time zero.

we

cyclones and also provide more consistent data. The examples used in this chapter are therefore dominated by Atlantic hurricanes making landfall in the USA.

In sections 4.2 and 4.3 ~~the approach is to~~ consider a single weather station associated with each cyclone, as close as possible to the landfall location, and we look at the single sea-level pressure and wind speed time series given at these stations. We consider 14 tropical cyclones selected for proximity of a weather station to the landfall location with reasonable data. Most

Small

of the cyclones selected, with the exceptions of Hurricane Jeanne and Hurricane Ernesto, are ranked category 4 or 5, using the SSHWS[4], at the time of landfall. These exceptions were included to broaden the scope of the study and because, although with a lower maximum wind speed, they exhibit a pronounced drop in pressure similar to the stronger cyclones. All 14 tropical cyclones are listed in table 4.1, and the sea-level pressure time series are shown in figure 4.1. Many more than these 14 were considered initially, but in many cases a HadISD weather station did not exist close to the landfall location or, when it did, the data were sparse.

we consider

In section 4.4 ~~we instead look at~~ the data given by several weather stations in a region surrounding the path of the cyclone. Because of better availability of data, all the cyclones in this case are Atlantic hurricanes making landfall on the Caribbean or Florida coastlines of the United States. The eight hurricanes considered are listed in table 4.2, Hurricane Andrew appears twice (labeled Andrew 1 and Andrew 2) because it made landfall twice: over Florida and then, two days later, over Louisiana. We therefore have a total of nine hurricanes. Again, we use the HadISD 2017 data set [Dunn et al., 2012, 2014, 2016; Smith et al., 2011]. We take

[4]The Saffir-Simpson hurricane wind scale (SSHWS) ranks cyclones based only on maximum one-minute sustained wind speeds. A cyclone with a wind speed of 209km/h or more is Category 4, with 251km/h or over is Category 5. (see www.nhc.noaa.gov/aboutsshws.php) ~~This is simply for ease of communication.~~

Name	Year	Location	Min pressure (hPa)	Max speed (km/h)	Osc. amp. (hPa)
Flo	1990	Shionomisaki (Japan)	890	270	0.91
Andrew	1992	Florida Keys (USA)	922	280	1.09
Opal	1995	Pensacola (USA)	916	240	1.22
Zeb	1998	Tuguegarao (Philippines)	900	285	1.15
Floyd	1999	Florida Keys (USA)	921	250	1.15
Charley	2004	Fort Myers (USA)	941	240	1.17
Frances	2004	Florida Keys (USA)	935	230	1.15
Jeanne	2004	Florida Keys (USA)	950	195	1.21
Katrina	2005	New Orleans (USA)	902	280	1.24
Rita	2005	Lake Charles (USA)	895	285	1.29
Ivan	2006	Mobile (USA)	910	270	1.18
Ernesto	2006	Florida Keys (USA)	985	120	1.05
Megi	2010	Tuguegarao (Philippines)	885	295	1.31
Hudhud	2014	Vishakhapatnam (India)	950	260	1.29

Table 4.1 The dates and locations of each of the 14 tropical cyclones considered. The "Location" column data is the identifying location recorded on the HadISD database for the weather station used in this study. The "Max speed" column gives the highest one-minute sustained wind speed, a value over 209km/h puts the cyclone into the ~~4 or~~ 5 SSHWS category. The data in the pressure and wind speed columns is not used in this study and is obtained from the NOAA archive (www.nhc.noaa.gov/data/#tcr) in the cases of the Atlantic hurricanes(those in the USA), and from www.wikipedia.org otherwise. The column "Osc. amp." gives the amplitudes of the 12-hourly oscillations in the sea level pressure data from that weather station but in a period 50 to 15 days before the cyclone event, this is ~~referenced~~ in section 4.1.3.

discussed

rather than 4

209~km/h tilde will give right space between number and unit and prevent line break

given for information only. It is obtained...

Name	Date	Region	Entry point
Andrew 1	24 August 1992	Florida	[25.4N, -79.0E]
Andrew 2	26 August 1992	Louisiana	[29.8N, -91.6E]
Katrina	29 August 2005	Louisiana	[29.3N, -89.6E]
Wilma	24 October 2005	Florida	[25.3N, -82.7E]
Gustav	1 September 2008	Louisiana	[29.3N, -90.8E]
Matthew	7 October 2016	Florida	[26.7N, -79.0E]
Harvey	25 August 2017	Texas	[25.2N, -94.6E]
Irma	10 September 2017	Florida	[24.5N, -81.5E]
Nate	8 October 2017	Louisiana	[29.3N, -89.2E]

selected

Table 4.2 The dates and locations of each of the nine hurricanes ~~considered~~. The entry point column gives the coordinates at which each hurricane entered the region, calculated using the HURDAT2 track data. Hurricane Andrew appears twice (labeled Andrew 1 and Andrew 2) because it made landfall twice: over Florida and then, two days later, over Louisiana.

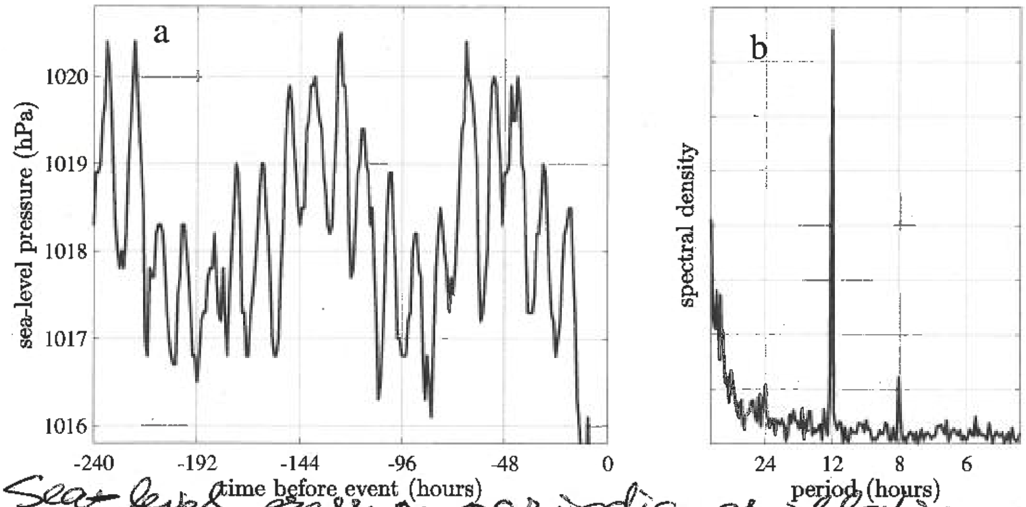
data from stations within 200km of the landfall of each hurricane and a region is formed by the minimal rectangle containing all of the weather stations used. We use the HURDAT2 track data ([Landsea and Franklin, 2013], see section 4.1.1) in each case to find the point at which the hurricane entered this region.

4.1.3 Filtering sea-level pressure oscillations

Upon inspection of the sea-level pressure data we see that there is a small, regular oscillation superimposed on the longer term fluctuations, figure 4.2a shows this phenomenon ~~in~~ at the Florida Keys station prior to Hurricane Andrew. An inspection of the power spectrum shows that the oscillations have a period of 12 hours (see figure 4.2b) and are due to the daily cycle. We calculate the mean amplitude of the oscillations over a given time period by subtracting the minimum pressure from the maximum pressure in a 12-hour sliding window and taking the mean. The "Osc. amp." column in table 4.1 records the mean amplitude found in the sea-level pressure data at each station in a period of 50 to 15 days before each of the tropical cyclone events: despite being recorded at different locations around the world over a period of 20 years, there is little variation in the size of these oscillations (mean: 1.17, variance: 0.011, range: [0.91, 1.31]). Because of this regularity, the possibility of removing the oscillations was proposed, and the effects of doing this were studied in the context of early warning signals, that is, we asked how the autocorrelation of the time series would be affected by the removal of the

tidal

swap



Sea level pressure periodic oscillations

Fig. 4.2 Panel **a**: A 240 hour segment of the sea-level pressure data preceding the Hurricane Andrew event, we note the 12-hour oscillations on the longer term fluctuations. Panel **b**: The periodogram obtained from a 35 day segment of the same time series (between 50 and 15 days before the event), we note the spike at 12 hours.

swap

oscillations. Two approaches were employed. First, a ~~deseasonalising~~ filtering approach where, given a time series $[Z]_k$ a deseasonalised series $[Z_\mathrm{D}]_k$ is created according to the formula

$$Z_\mathrm{D}(k) = Z(k) - \underset{i=k \bmod 12}{\mathrm{mean}}\,(Z(i)). \tag{4.1}$$

The other approach is to subtract a sine wave with 12 hour period, that is, a new series $[Z_\mathrm{S}]_k$ is given according to

$$Z_\mathrm{S}(k) = Z(k) - A\sin\left(\frac{2\pi}{12}(k-\phi)\right) \tag{4.2}$$

where A is the mean amplitude found over the entire series, as described above, and ϕ is required to align the series Z with the sine wave. In practice, the phase ϕ was found in each case by minimising the variance of Z_S over values of $\phi \in [0,12)$.

Both of these approaches are applied to pressure data from the Florida Keys station in the period of 50 to 15 days before the appearance of Hurricane Andrew, and for both of the new series, as well as the original raw series, the ACF1 indicator is calculated using a window size of 100 points. The results are shown in figure 4.3. We note that both methods of removing oscillations give almost identical results, both qualitatively and when considering the autocorrelation of the resulting series. The effect on the ACF1 indicator is, effectively, of smoothing out the larger fluctuations but does not appear to change the general shape of the indicator series (that is, longer scale increases or decreases). For this reason the 12-hour oscillations in pressure were not removed ~~by these methods~~ in further analysis of the tropical cyclone data.

4.2 Application of one-dimensional tipping point techniques

In this section we consider the time series of the sea-level pressure variable at a fixed location in the time immediately before the arrival of a tropical cyclone in that location. We use the fourteen tropical cyclones that are detailed in table 4.1 (see section 4.1.2).

In the case of each of the fourteen cyclones, the sea-level pressure time series is centred on the minimum pressure value which is labelled $t = 0$. This cannot be called analogous to the bifurcation point in the pitchfork bifurcation example of chapter 2 (figure 2.8) but is a common feature of all time series which is convenient to use for the purpose of comparison.

4.2.1 Method

The same methods are applied to the sea-level pressure time series here as are applied to the model examples in section 2.4. The ACF1, DFA and PS indicators (definitions 2.3.1, 2.3.2 and 2.3.3 respectively) are applied to the times series with a sliding window of approximately 100 points, equivalent to 100 hours in this data. The exact window sizes used, for each indicator, are: ACF1: 90points; DFA:90points; PS:102points. These values are determined by the sensitivity analysis which will be explained in section 4.2.3.

~~For~~ Each of the ~~three~~ ~~indicators, the~~ indicator series is calculated for each of the fourteen time series ~~and the mean over these fourteen series is calculated, with error bars of one standard deviation.~~

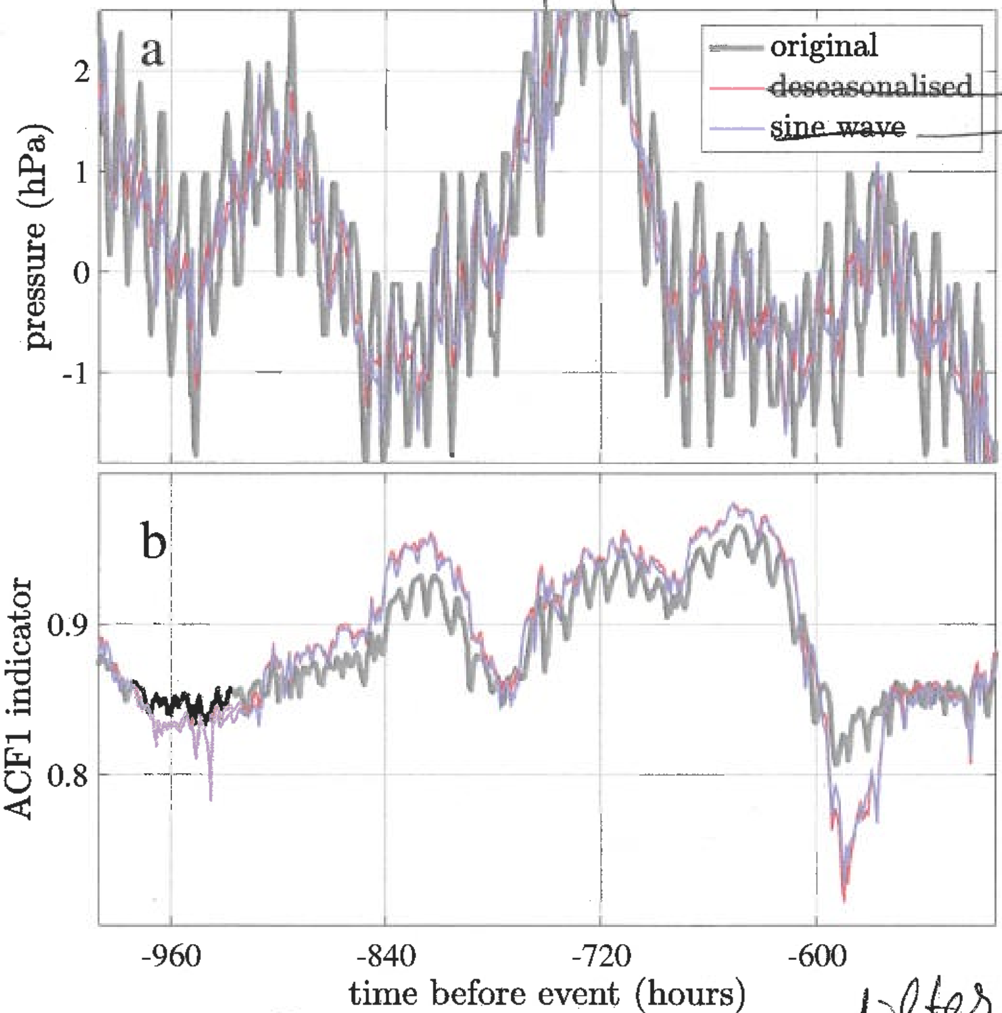
↓ repeated below

4.2.2 Results

Figure 4.4 shows the fourteen sea-level pressure time series centred at the point of minimum pressure (panel **a**) and the mean of the three indicator series (ACF1, DFA and PS indicators in panels **a**, **b** and **c**). The mean indicator series are shown with error bars of one standard deviation.

We note that the ACF1 indicator is consistently high, greater than 0.9, over the entire series and does not apparently rise as a precursor of the cyclone event. The mean of the DFA indicator increases within the time period (24 hours before the event) that the drop in pressure is obvious in the time series, and so the increase cannot be said to be an early warning signal.

?



filtered
removed sine data

filtering(4.1)

Fig. 4.3 Three hourly sea-level pressure time series from the Florida Keys station in a period from 50 to 15 days before Hurricane Andrew: the original raw data; the same data ~~with~~ oscillations removed by ~~deseasonalising;~~ and the data with oscillations removed by subtracting a sine wave. Panel **a** shows the pressure series (mean-centred for clarity) whilst panel **b** shows the ACF1 indicator of the series with a 100 point window.

Eq. (4.2)

The PS indicator, however, does show an increasing trend starting at around 50 hours before the minimum pressure, although it is not significant at the level of one standard deviation.

due to the noisy

actual periodicity

4.2.3 Sensitivity analysis

In order to calculate the indicator series shown in figure 4.4, it is necessary to select an appropriate window size. The calculation of each of the indicator series requires an appropriately large window size in order to yield an accurate calculation of the scaling exponent. The calculation of the power spectrum scaling exponent from the periodogram obtained using the fast Fourier transform is particularly inaccurate when only a small number of points are available.

V √ V Besides the error ~~present in the~~ periodogram, ~~with fewer points~~ the spikes in the periodogram, such as the spike representing the 12-hour oscillations, have a greater influence on the calculation of the scaling exponent. Furthermore, the calculation of the DFA exponent requires V √ a ~~minimum~~ number of points depending on the parameters selected. The lag-1 autocorrelation could reasonably be calculated for a series of only three points, but the ACF1 indicator exhibits large variability, as with the PS and DFA indicators, for small window sizes.

certain

It is therefore desirable to use a ~~very~~ large number of points to give an accurate estimate of the scaling exponent with ~~less~~ variability ~~over time~~. However, one must attain a compromise against the competing requirement for a short time window corresponding to the time scale of the phenomenon under investigation. For example, when we consider the sea-level pressure time series in figure 4.4 we note that the visible influence of the cyclone on the pressure begins only around 24 hours before the minimum. Attempting to detect an early warning signal an sufficiently indicator series calculated with a window of several weeks would clearly not be useful: the large number of points relative to the actual period of interest will smooth out any significant increase or decrease. There is also a limit imposed by the length of the time series data. We require a window size such that the indicator variance is significantly small that we can see the noisy temporal EWS, but that the rise in the indicator value before the tipping event is large enough so that we can recognise it as an EWS.

Here we analyse the sensitivity of each indicator to the window size used, in order to ascertain the optimal window size in this particular example of a physical phenomenon, if such a universal optimum exists.

We expect that the EWS apparently becomes stronger with decreasing window size, above a certain size where the variability becomes so large that it is not possible to detect an EWS. In this way, we make a subjective choice as to the optimum window size to use in this context. The hypothesis is that all cases considered here (all fourteen cyclones and their corresponding sea-level pressure time series) are sufficiently similar in the way that the scaling exponents behave that a general optimum (or near-optimum) window size exists. In this case, it will be

30

possible to apply these indicator methods to new data from similar sources ~~without further analysis.~~ .

It is useful to produce a contour plot of the mean value of the indicator over time, with window-size on the y-axis. In this way we are able to visualise hundreds of indicator series simultaneously. present Calculated

In figure 4.5 we ~~have calculated~~ the PS indicator series for window sizes from 40 points to 160 points. We see that there is an increasing trend in the PS indicator towards the end of the time series with almost all window sizes. The exceptions occur periodically where the window size is a multiple of twelve, using these window sizes the indicator has a high value (≈ 1.5) over the entire series. The most obvious early warning signals occur with window sizes which are an odd multiple of 6, that is, 90, 102, 114, etc.. Using larger values (e.g. 174) the useful variability in the indicator is smoothed out so that the trend towards the end of the series is less obvious. We hypothesise that this 12-hour pattern is created by the 12-hour oscillations present in the sea-level pressure data, which particularly influences the power spectrum (the pattern is not observed in the ACF1 and DFA indicators, see figures 4.8 and 4.9). We also observe that there is a qualitative change in the nature of the pattern as the window size is increased from 99 points to 100 points~~, neither of which numbers are multiples of six. At the~~ at which point ~~interval between 99 and 100~~ on the y-axis, all of the contour lines are lined up horizontally. This observation has been mentioned to the team behind the HadISD dataset, namely the authors of Dunn et al. [2016], and it has been ruled out that this is an artefact of the data collection or filtering. The investigation of the cause of this effect may be a topic for future work involving the PS indicator applied to sea-level pressure data. The 12-hour pattern serves to highlight to necessity of considering ~~the~~ such sensitivity of the method to the window size parameter ~~, as we have done here~~ in future studies. forewarning

We hypothesise that a time scale of about 100 hours, or four days, is a cut-off for the study of tropical cyclone in the context of early warning signals. A longer window encompasses too much of the time series before any signal is created, that is, before the cyclone has any influence on the autocorrelation of the sea-level pressure. Since the data available is hourly, this promotes an unfortunately short window of 100 points or fewer, which introduces a large amount of variability into the indicator series due to noise-sensitivity in the scaling exponent calculations. For this reason we proceed, in sections 4.3 and 4.4 to incorporate more data into the calculation of a tipping point indicator series.

In keeping with the decision to select a window size of about 100 point, we selected a window size of 102 points for when calculating the PS indicator in figure 4.4. At 90 points, the indication does not provide such an obvious early warning signal. In figure 4.6 we present the PS indicator, as in figure 4.4d, but using window sizes of 54, 90, 102 and 150 points (in along the time scale

panels **a**, **b**, **c** and **d** resp.). Since the fast Fourier transform periodogram is a less accurate approximation of the power spectrum when using fewer points, we might expect there to be greater variability when using a short window: both less agreement between the fourteen time series, and more variability within each individual time series. We observe, in figure 4.6, that the former is not apparently the case: the size of the error bars, which is one standard deviation over the fourteen sea-level pressure series, is largest when using a window size of 102 points, which is our chosen value based on the more obvious early warning signal in the mean.

In figure 4.7 we investigate in what way the variance in the indicator series is effected by the window size. We expect, since the estimation of the DFA amd PS scaling exponents is less accurate for shorter time series, that using a shorter window would result in a noisier indicator series, regardless of the nature of the input data. For each window size we calculate the standard deviation in the indicator series for each of the fourteen tropical cyclones, then take the mean of the fourteen standard deviations. The indicator series is calculated between 300 hours and 48 hours before the event, so that any ~~desirable or~~ anticipated increase prior to the event (in the 48 to 0 hours range) is not confused with noise. The hypothesis proves correct presumably for the DFA indicator, shown in red, until the window size reaches around 100 points, after which the standard deviation remains constant. For the PS indicator, the standard deviation also decreases with increasing window size, as expected, but jumps up again for window size 100 points, after which it continues to decrease steadily. This jump, ~~we presume~~, is related to the qualitative change in the mean which also occurs at window size 100 points (see figure 4.5). As we have remarked above, it is not known why this occurs but the investigation of the effect is an interesting topic of future work.

The PS indicator series is less noisy for window size 90 points, and also has less variability across the fourteen cyclones, than for 102 points. However, this must be balanced by the consideration that the early warning signal, if present at all, is more obvious in the 102 point window series.

Figure 4.8 is produced by the same method as figure 4.5 but uses the ACF1 indicator. The range of window sizes is also smaller: from 40 to 120, because for sizes over 120 the indicator was consistently above 0.95 and showed no change over the time period. In this case, we notice that the indicator rises, for all window sizes, to a maximum value at around 16 hours before the event, before decreasing sharply. Unfortunately there is something just before -100 hours which causes the ACF1 indicator, particularly with a window size above 60, to be very high –higher than the highest value preceding the cyclone event– which is not visible in the PS and DFA indicators (figures 4.5 and 4.9).

Figure 4.9 is produced using the DFA indicator, again with a range of window sizes from 40 to 120. The calculation of the DFA scaling exponent is not possible for very short series

ignore

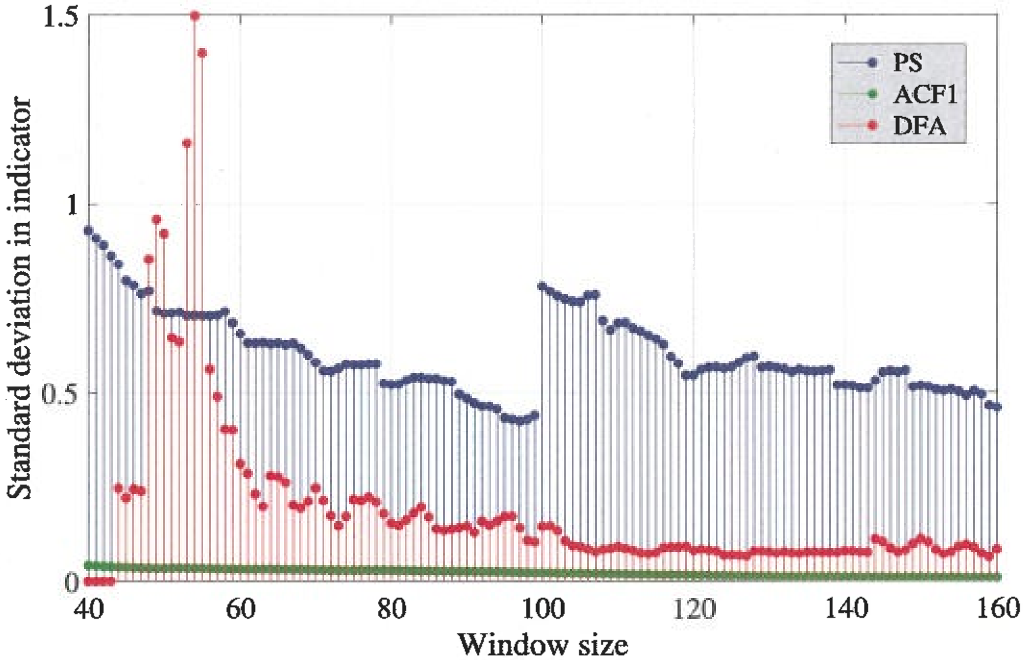
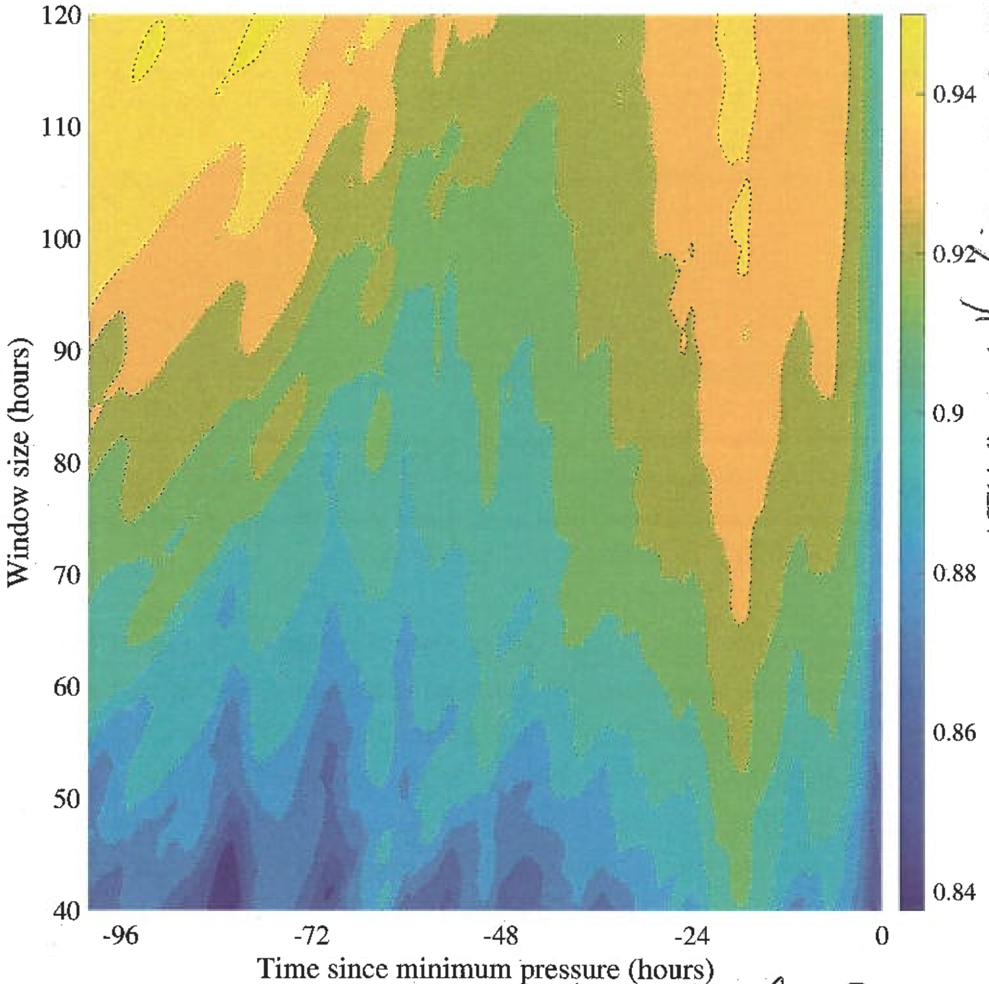


Fig. 4.7 The PS indicator is calculated as in figure 4.4d, but using window sizes 54, 90, 102 and 150 points (in panels **a**, **b**, **c** and **d** resp.).

depending on how the series is segmented before each segmented is detrended by subtracting a polynomial[5]. So as to be able to estimate the scaling exponent, we ~~here~~ use eight different segment sizes increasing logarithmically from the smallest to the largest, the smallest being ten points and the largest being $\lfloor N/4 \rfloor$ where N is the length of the series. In this way it is always possible to get at least four segments from the series. Using this method it does not make sense to calculate the DFA exponent with a series shorter than 44 points because only one distinct segment size is used, 10, and therefore the relationship between the segment size and the fluctuations coefficient is estimated from a single point.

Table 4.3 shows the eight segment sizes used for series lengths 40, 54, 100 and 400. A series length of at least 400 would be ideal, giving a range of segment sizes between 10 and 100 with which to be able to calculate the correlation between the segment size and fluctuations coefficient (see section 2.1.2). With the available data for the sea-level pressure, a window size of 400 points cannot be used. We use a window size of 90 points because there appears to be a slightly stronger early warning signal (an increase in the indicator prior to the event) than with other window sizes. However, the DFA indicator is practically useless when applied to this data.

[5]In this study we consistently use quadratic detrending. to address nonlinear effects.



What is measured, Kendall?

Kendall? Actual value?

Fig. 4.8 Mean ACF1 indicator for tropical cyclone data (see fig. 4.4) calculated using window sizes from 40 to 120. The ACF1 indicator appears to rise around 40 hours before the cyclone event in almost all cases, the exceptions occur when the window size is a multiple of 12, in these cases the indicator is high (greater than 1) over the entire series.

ACF cannot exceed 1, it is normalised

Series length	Segment sizes
40	10
54	10, 11, 12
100	10, 11, 12, 14, 16, 19, 21, 25
400	10, 13, 19, 26, 37, 51, 71, 100

Table 4.3 During the calculation of the DFA scaling exponent, a series of length N is segmented eight times, the segment sizes increase logarithmically from 10 to $\lfloor N/4 \rfloor$. The segment sizes used for four example series lengths are shown. For very short series, fewer than eight different segment sizes may be used.

nonoverlapping segments? Say it then.

did you try overlapping segments?

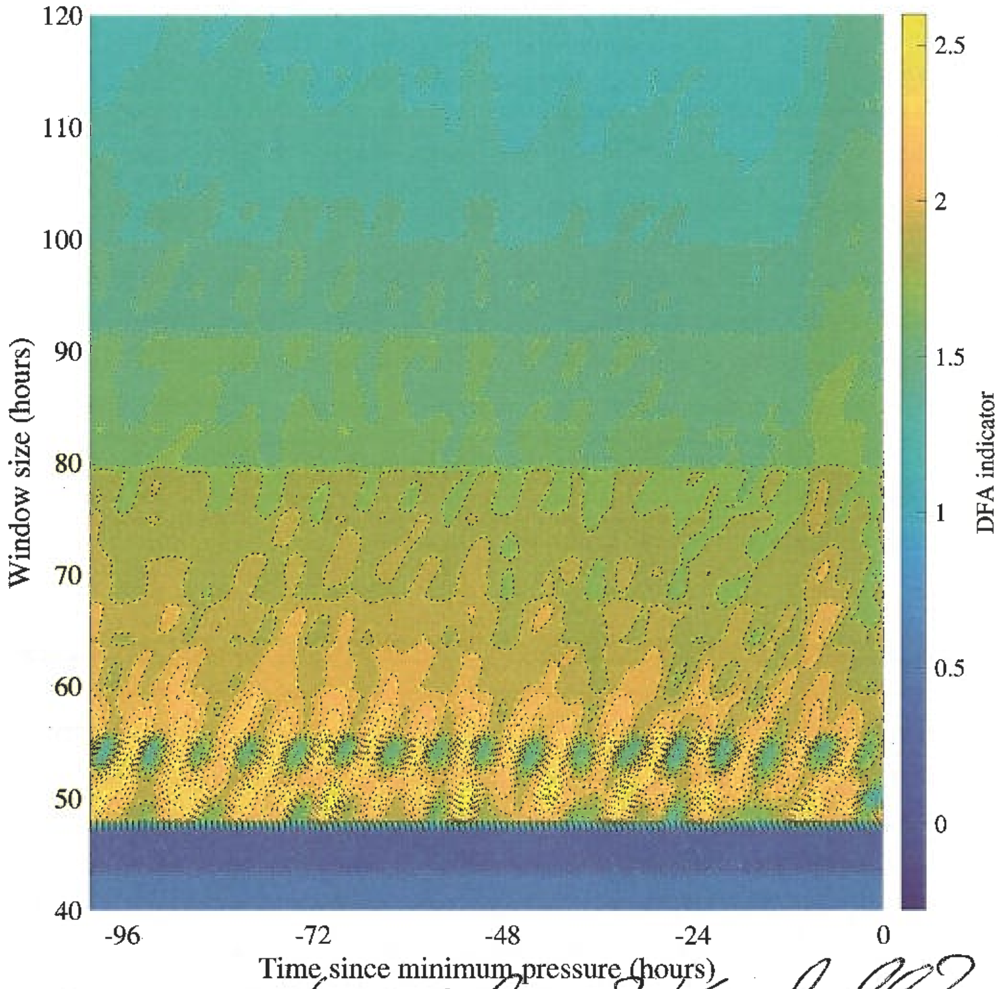
A series length of 54 is considered in table 4.3 because there appears to be interesting oscillating behaviour in figure 4.9 with a window size of 54. The oscillation has a period of six hours, which suggests a connection with the 12-hour oscillations present in the sea-level pressure data. We also note that there is a qualitative change in the behaviour of the DFA indicator as the window size reaches 80, 91 and (more weakly) 100 points, similar to the change observed in the PS indicator at 100 points (see figure 4.5). Again, it is not known why this occurs and should be the subject of further investigation.

EWS

4.3 Application of multi-variable tipping point techniques

In this section we consider the same fourteen tropical cyclones ~~as in section 4.2,~~ which are detailed in table 4.1. In section 4.2, ~~tipping point~~ indicators were applied to the one-dimensional time series of the sea-level pressure variable at each station. Here, both the wind speed and sea-level pressure are considered together. We first calculate the ACF1, DFA and PS indicator series for the wind speed variable. Using the exact same process by which figure 4.4 was produced, showing the indicator series for the sea-level pressure variable, we produce figure 4.10, showing the indicator series for the wind speed variable. The same window sizes were used for the ACF1, DFA and PS indicators (90, 90, 102 resp.) as for the sea-level pressure. We note that the mean ACF1 and PS indicators rise with the rising wind speed, and that the DFA indicator performs similarly as in the sea-level pressure case. None of the indicators appear to provide an early warning signal, so we do not expect that combining the wind speed and sea-level pressure variables will yield a signal stronger than already present in the sea-level pressure. However, we proceed to investigate the results of the multi-variable tipping point analysis in comparison to the analysis of the individual variables.

Throughout this section, including in figure 4.10, the time series are aligned so that $t = 0$ occurs at the point where the sea-level pressure is minimum, for consistency when comparing



actual values? Kendall?

Fig. 4.9 Mean DFA indicator for tropical cyclone data (see fig. 4.4) calculated using window sizes from 40 to 120. The DFA indicator appears to rise around 40 hours before the cyclone event in almost all cases, the exceptions occur when the window size is a multiple of 12, in these cases the indicator is high (greater than 1) over the entire series.

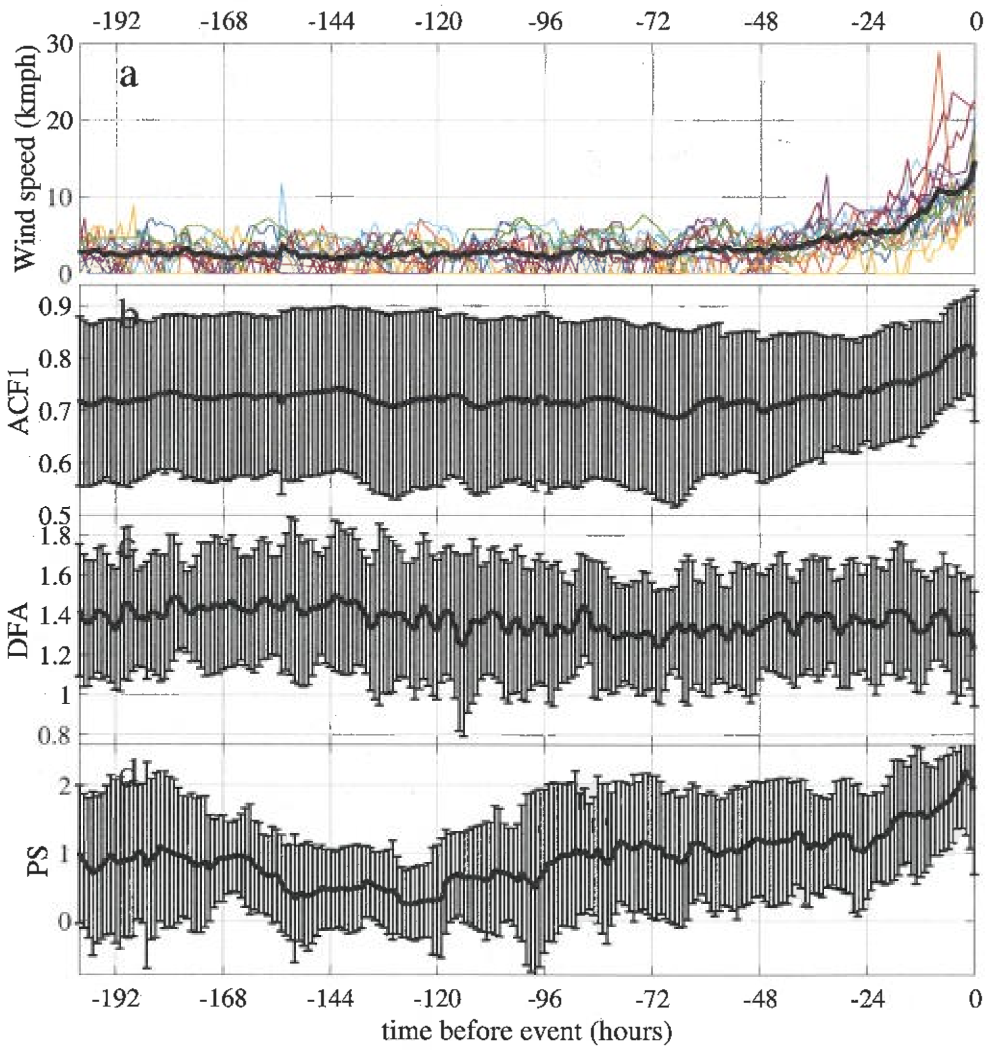
to the results in the previous section. This may not be the same point at which wind speed is maximum.

4.3.1 Method

Having considered the sea-level pressure and wind speed variables separately (figures 4.4 and 4.10) we now ~~now~~ combine the two variables so that more data is available from which to produce a single ~~tipping point~~ EWS indicator series. We use the ~~the~~ methods introduced in sections 3.1 and 3.2, that is, the analysis of the eigenvalues of the Jacobian matrix of a linearised system, and the use of EOFs to reduce the dimension before applying the one-dimensional indicator techniques.

For the Jacobian eigenvalues analysis we repeat the method in section 3.1 using a window size of 102 points in order to provide a comparison with the PS indicator applied to the sea-level pressure data, for which this window size is used. In section 4.3.3 we will perform a sensitivity analysis to determine whether a different window size is preferable. Besides the window size, it is necessary to choose whether to study the real part, the imaginary part, or the complete complex eigenvalue. Since we do not know in advance the nature of the tipping, we record both the real and imaginary parts. Thus, in a sliding window of 102 points, the Jacobian matrix of the linearised system is estimated from the two system variables (sea-level pressure and wind speed) and the principal eigenvalue is recorded. The method of approximating Jacobian eigenvalues has previously been applied to dynamical systems where the governing equations and the type of bifurcation are known (Williamson and Lenton [2015]) and so we know in advance what type of signal we should expect, such as with the example of the Van der Pol oscillator in the previous chapter (figure 3.6). However, we have noted that the method was intentioned as a two-dimensional analogue to the lag-1 autocorrelation, so it is reasonable to apply it to systems in which the one-dimensional ACF1 indicator, and the related PS and DFA indicators, increase prior to a tipping point. We are particularly interested to note the extent to which this indicator is an improvement, if at all, over simply considering the one-dimensional indicators individually, and whether any additional artefacts arise as a consequence of considering the two variables together.

The other technique we use here is to first obtain the the leading EOF score of the two time series (sea-level pressure and wind speed) in order to reduce the dimension to one. The time series in both variables are first mean-centred, as per the EOF method, and also normalised in order to non-dimensionalise. We are then able to apply the one-dimensional EWS techniques, the ACF1 and PS indicators, and ~~then~~ take the mean over all cyclones. We use a window size of 102 for the PS indicator and 90 for the ACF1 indicator, since these were found to be useful when considering the sea-level pressure alone in the previous section.



Wind speed datasets and their

Fig. 4.10 ACF1, DFA and PS indicators ~~applied to wind speed data~~, mean shown in black. (a) Data from the 14 tropical cyclones. (b,c,d) The mean ACF1, DFA and PS indicators, with error bars of 1 standard deviation.

As noted in section 3.2.2, there are three available approaches to obtaining the principal EOF score, that is, there are three approaches to obtaining the vector onto which we project the two-variable series. The vector is the principal eigenvector of the covariance matrix of two time series, we may use either

1. the entire available time series (global projection), or
2. only the segment of the time series in each window (windowed projection), or
3. the entire available time series *up to* the end of the current window (moving projection).

The first option in this case implies using the part of the time series which lies in the "future" of the current window. This is not comparable to practical prediction problems and so we avoid it here. It is also not sensible to obtain the projection vector using the part of the time series which includes the tipping point when considering the rest of the series, because the dynamics are not representative.

The second and third options are distinguished in section 3.3.2. The windowed projection most accurately captures the maximum variance in each window, but may be unsuitable if the principal EOF eigenvector is highly variable due to noise. In this ~~our~~ system, we want to avoid considering signals from meteorological events prior to the tropical cyclone being studied and want to ~~, as much as possible~~, consider a relatively stationary system prior to the tipping event. To this end, we generally truncate the time series at around 300 hours before the tipping point and so the total length of the series is not much longer than the window size used in the indicator calculation (approximately 100 hours). Therefore, the difference between using the windowed or moving projection ~~will~~ would be negligible. Throughout this section, we use the windowed projection.

good writing here

Since the two variables have different units, it is not meaningful to consider them together without first non-dimensionalising. To satisfy this requirement, as part of the EOF technique, each time series is mean-centred and normalised by its own standard deviation. In section 4.3.3 we explain the investigation into the effect of weighting the two variables differently, as a result of which we choose to weight the sea-level pressure and wind speed variables in the ration $40:60$ when using the ACF1 indicator, and to weight the variables equally when using the PS indicator.

4.3.2 Results

The Jacobian eigenvalue analysis is performed separately for each cyclone and both the real and imaginary parts of the principal eigenvalue are recorded. The mean of the fourteen resulting indicator series is shown in figure 4.11a,b, with error bars of one standard deviation over

Mann-Kendall of

Data	Indicator	
	ACF1	PS
Sea-level pressure	-0.20	0.79
Wind speed	0.91	0.70
EOF	0.45	0.72

and

Table 4.4 Comparison of the gradient of the ACF1 ~~abd PS~~ indicators of the time series data in a 30-hour window before the tropical cyclone event. We compare sea-level pressure and wind speed data alone (as presented in figures 4.4 and 4.10) to the EOF score of the two series (figure 4.11c,d). In the case of each indicator, the EOF result has a gradient somewhere between the two considered alone.

the fourteen indicators. We note that there is a slight increasing trend in the real part of the eigenvalue starting at around 48 hours before the minimum pressure, similarly to the PS indicator applied to the sea-level pressure series (figure 4.4). The imaginary part also exhibits a spike as the event occurs.

The results of applying the ACF1 and PS indicators to the windowed EOF projection are shown in figure 4.11c,d. We notice that the rise in the ACF1 indicator is more noticeable than when using only the sea-level pressure (figure 4.4b), but is not so noticeable as when using only the wind speed (figure 4.10b). The combination of the two variables using EOF has given an indicator series which appears to be somewhere between the two separate indicator series, although it is not equivalent to simply taking the mean of the two and does give more weight to the more visible indicator. The PS indicator series is more similar to the PS indicator series using only sea-level pressure (figure 4.4d). We quantify the gradient of the indicator by evaluating the Mann-Kendall coefficient of the mean indicator series in the 30-hour window before time zero, using the equation

$$\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \operatorname{sign}(X_j - X_i), \tag{4.3}$$

where X is the time series of the indicator. The result is summarised in table 4.4.

4.3.3 Weighting sensitivity in EOF techniques

many

The Jacobian eigenvalues technique takes into account only the covariance between the two variables and is not sensitive to the relative weights of the variables. That is, if the data in one column were changed to different units resulting in values ~~1000~~ times larger, the eigenvalues (but not the eigenvectors, which are not studied) would not change. The calculation of EOFs, however, is sensitive to different weighting schemes. By multiplying each variable (each column of the data matrix) by different scalar values, we are able to assign greater or lesser importance to each variable.

We are therefore able, by visualising the indicator series for a variety of different weighting schemes, to select an optimum weighting, in each case. For the data matrix X, where the columns of X are the time series variables, we instead use the weighted matrix

$$X_w = XW, \tag{4.4}$$

where, in our case,

$$W = \begin{bmatrix} p & 0 \\ 0 & 1-p \end{bmatrix}, \quad p \in [0,1]. \tag{4.5}$$

For $p = 0$ the first column of X, in our case the sea-level pressure variable, is given no weight and the indicator series is identical to the one-dimensional indicator applied to wind speed alone. Conversely, when $p = 1$, only the sea-level pressure is considered.

The ~~widowed~~ EOF projecting is found for a range of values from $p = 0$ to $p = 1$ and the PS and ACF1 indicators are calculated. The results for the ACF1 indicator and the PS indicator are both shown as contour plots in figure 4.12. We note that, when using the ACF1 indicator, there is a clear difference between $p < 0.5$ and $p > 0.5$. Weighting the sea-level pressure (SLP) variable at 0.4 is effectively equivalent to weighting it zero. Likewise, weighting the SLP variable at 0.6 is effectively equivalent to weighting it 1. When SLP is weighted more heavily than wind speed ($p > 0.5$) the ACF1 indicator is consistently high and does not provide an EWS, which is already apparent from the ACF1 indicator applied to SLP alone (see figure 4.4b). Using $p < 0.5$ there is an increase in the indicator prior to the event, as in figure 4.10b, and this rise appears earlier as the relative weight of the SLP variable increases, up to the complete qualitative change at $p = 0.5$. It appears that choosing a value $p = 0.4$ would provide the best EWS in this case.

For the PS indicator, the picture is more similar for different weighting schemes. The contour plot is complicated by a few unexplained spikes with much higher indicator values than the surrounding points. For example, for the specific weighting $p = 0.15$ there occurs a spike at -12 hours where the PS indicator is greater than 3. It is possible that this approach could be used to determine specific weightings to avoid, based on the presence of spikes, but this would not be practical without prior knowledge of the spikes' distribution.

max value?

which may be caused by tidal periodicity spike

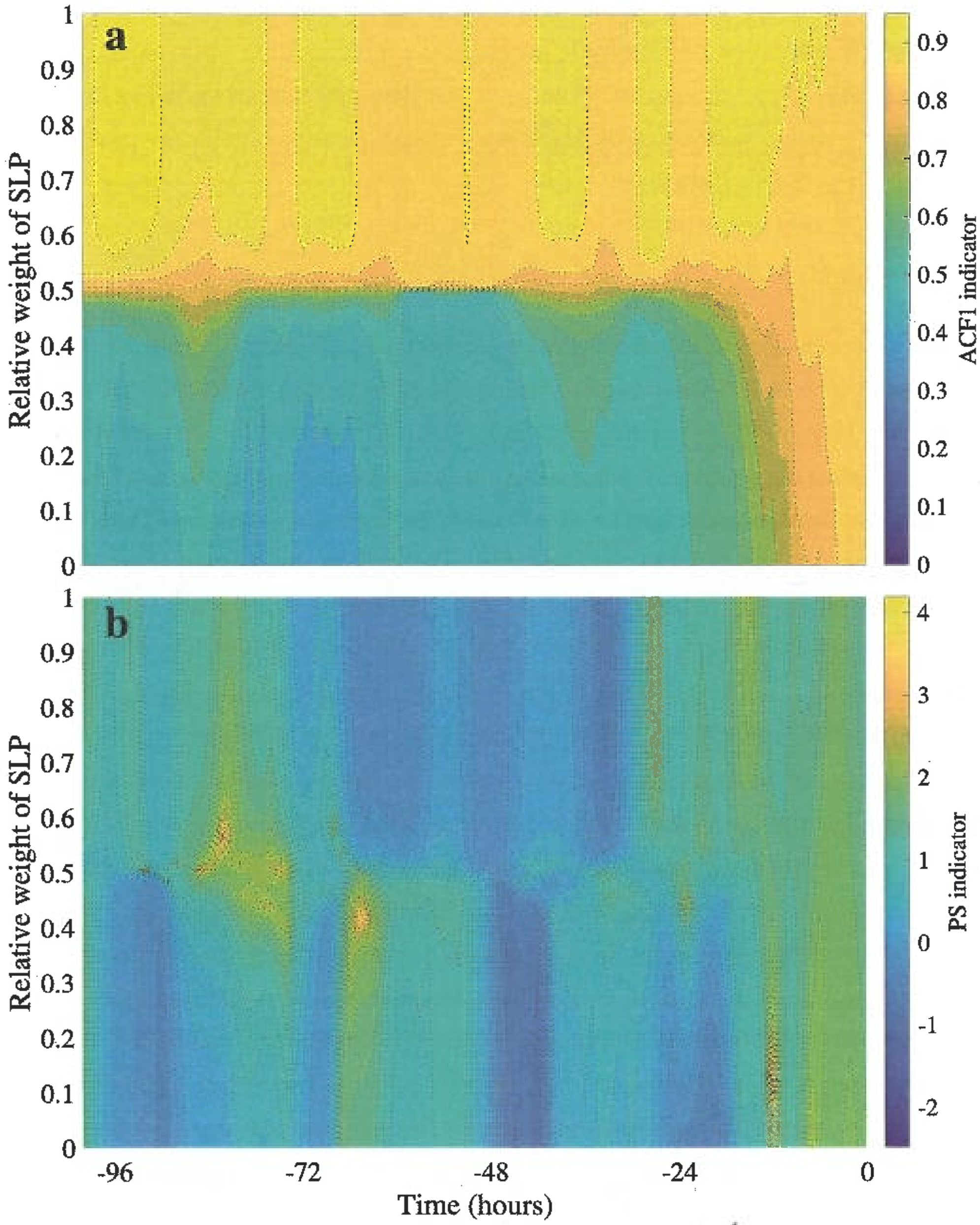


Fig. 4.12 Caption

distributed

4.4 Early warning signals for tropical cyclones using spatially separated data

Here we make an attempt to use the one-dimensional EWS indicators applied to multiple time series data distributed over a geographic area. ~~By using more data we expect to improve the performance of the EWS techniques.~~ We ~~also~~ investigate the spatial variation in the indicator series, potentially with a view towards detecting a spatial pattern which may provide a warning of the arrival of the tropical cyclone.

4.4.1 Method

For this analysis nine separate tropical cyclone events were selected (see section 4.1.2) which are described in table 4.2. These events were selected due to an availability of data at several stations within 200km of the landfall location of the hurricane. The sea-level pressure data were obtained from each station within this radius in a period of twenty days before landfall, and stations were disregarded where > 10 of points were missing. The data from the remaining stations were linearly interpolated onto the entire one-hourly time series. The ?

For each of the nine hurricanes, we take the sea level pressure data from ~~all~~ available stations in the region supplied by the HadISD 2017 dataset. We then calculate the PS indicator series for each pressure ~~series~~ dataset and asses the slope of that indicator series, using the Mann-Kendall coefficient (see equation 4.3), in a window of approximately 30 hours before the hurricane enters the region. If the behaviour here is similar to the behaviour of the mean of the PS indicator applied to the individual sea level pressure series (figure 4.4d), we would expect that an upward trend in the indicator (i.e. a high value of the Mann-Kendall coefficient) will be detectable for each location within 30-hours travelling time of the hurricane. We expect that the slope of the PS indicator series would be high at points where the hurricane is very close and lower at points further from the point where the hurricane enters the region.

~~The slope of the PS indicator is assessed, in each case, using the Mann-Kendall coefficient given in equation 4.3, following Lenton et al. [2012] who use the Kendall statistic as a measure of the increase in the ACF1 and DFA indicators.~~ The window size used in the calculation of the PS indicator series, and the length of the window in which the Mann-Kedall coefficient is evaluated, are determined for each hurricane through a sensitivity analysis similarly to the process described in section 4.2.3, which is presented in figure 4.13. In these contour plots, the mean is not taken over the indicator series for all tropical cyclones, as in section 4.2.3, but rather an individual contour plot is presented for each cyclone in which the mean is taken over all the stations in that region. We see that there is again a significant qualitative change as the

positive

distributed

4.4 Early warning signals for tropical cyclones using spatially separated data

Here we make an attempt to use the one-dimensional EWS indicators applied to multiple time series data distributed over a geographic area. ~~By using more data we expect to improve the performance of the EWS techniques.~~ We ~~also~~ investigate the spatial variation in the indicator series, potentially with a view towards detecting a spatial pattern which may provide a warning of the arrival of the tropical cyclone.

4.4.1 Method

For this analysis nine separate tropical cyclone events were selected (see section 4.1.2) which are described in table 4.2. These events were selected due to an availability of data at several stations within 200km of the landfall location of the hurricane. The sea-level pressure data were obtained from each station within this radius in a period of twenty days before landfall, and stations were disregarded where > 10 of points were missing. The data from the remaining stations were linearly interpolated onto the entire one-hourly time series. The ?

For each of the nine hurricanes, we take the sea level pressure data from ~~all~~ available stations in the region supplied by the HadISD 2017 dataset. We then calculate the PS indicator series for each pressure ~~series~~ dataset and asses the slope of that indicator series, using the Mann-Kendall coefficient (see equation 4.3), in a window of approximately 30 hours before the hurricane enters the region. If the behaviour here is similar to the behaviour of the mean of the PS indicator applied to the individual sea level pressure series (figure 4.4d), we would expect that an upward trend in the indicator (i.e. a high value of the Mann-Kendall coefficient) will be detectable for each location within 30-hours travelling time of the hurricane. We expect that the slope of the PS indicator series would be high at points where the hurricane is very close and lower at points further from the point where the hurricane enters the region.

~~The slope of the PS indicator is assessed, in each case, using the Mann-Kendall coefficient given in equation 4.3, following Lenton et al. [2012] who use the Kendall statistic as a measure of the increase in the ACF1 and DFA indicators.~~ The window size used in the calculation of the PS indicator series, and the length of the window in which the Mann-Kedall coefficient is evaluated, are determined for each hurricane through a sensitivity analysis similarly to the process described in section 4.2.3, which is presented in figure 4.13. In these contour plots, the mean is not taken over the indicator series for all tropical cyclones, as in section 4.2.3, but rather an individual contour plot is presented for each cyclone in which the mean is taken over all the stations in that region. We see that there is again a significant qualitative change as the

positive

when

window size increases over 100 hours. We note that the increase in the indicator is usually seen, in a 30-50 hour window before the event, which is consistent with our expectations based on previous analysis.

As a result of the sensitivity analysis in figure 4.13 we take the decision to measure the Mann-Kendall coefficient of the indicator series (for the time series at each spatial location, separately) in a 30-50 hour window before a time T_{entry}, which is ~~the time at which~~ the cyclone enters the region under study: that is the time at which the minimum pressure is attained at the first spatial location on the cyclone's path. We expect to see a high coefficient value (i.e. a large increase in the indicator) at locations close to the entry point. Also, according to the sensitivity analysis, we note that the PS indicator is most useful when using a window size which is an odd multiple of six, consistent with the result of the ~~similar,~~ previous sensitivity analysis in section 4.2.3. This is likely due to the 12-hour oscillations in the data. For each cyclone we have chosen an indicator window size of either 90 or 102 points ~~based on inspection~~.

tidal

4.4.2 Results

The value of the Mann-Kendall coefficient of the PS indicator is plotted as a filled contour plot over the geographic area. In figure 4.14 we present a summary of the results: in each case, the contour map is rotated so that the forward path of the Hurricane is moving from the bottom to the top of the image. Hurricane tracks, obtained from the NOAA HURDAT2 dataset (see section 4.1.1) are shown by the black line. We are generally able to see a movement of the hurricane from areas with a high value of the Mann-Kendall coefficient, towards areas with lower value, as we expect. Whilst this doesn't provide a useful EWS, it is clear that the PS indicator ~~is more robust~~ provides a more robust signal when applied to many time series in a region than when applied to individual series, as in figure 4.4d.

4.5 ~~Development of a~~ A model of sea-level pressure in tropical cyclones

Holland [1980] presents a simple analytic model of the pressure (p) profile in hurricanes:

$$p(r) = p_c + (p_n - p_c) \exp\left(\frac{-A}{r^B}\right), \tag{4.6}$$

where r is the radial distance from the centre of the hurricane, p_c and p_n are the central and ambient pressures, A and B are parameters to be determined by fitting to observed data. Holland [1980] fits the model to three Australian hurricanes: Tracy (December 1974), Joan (December 1975) and Kerry (February 1979). The values of parameters A and B are obtained by fitting to

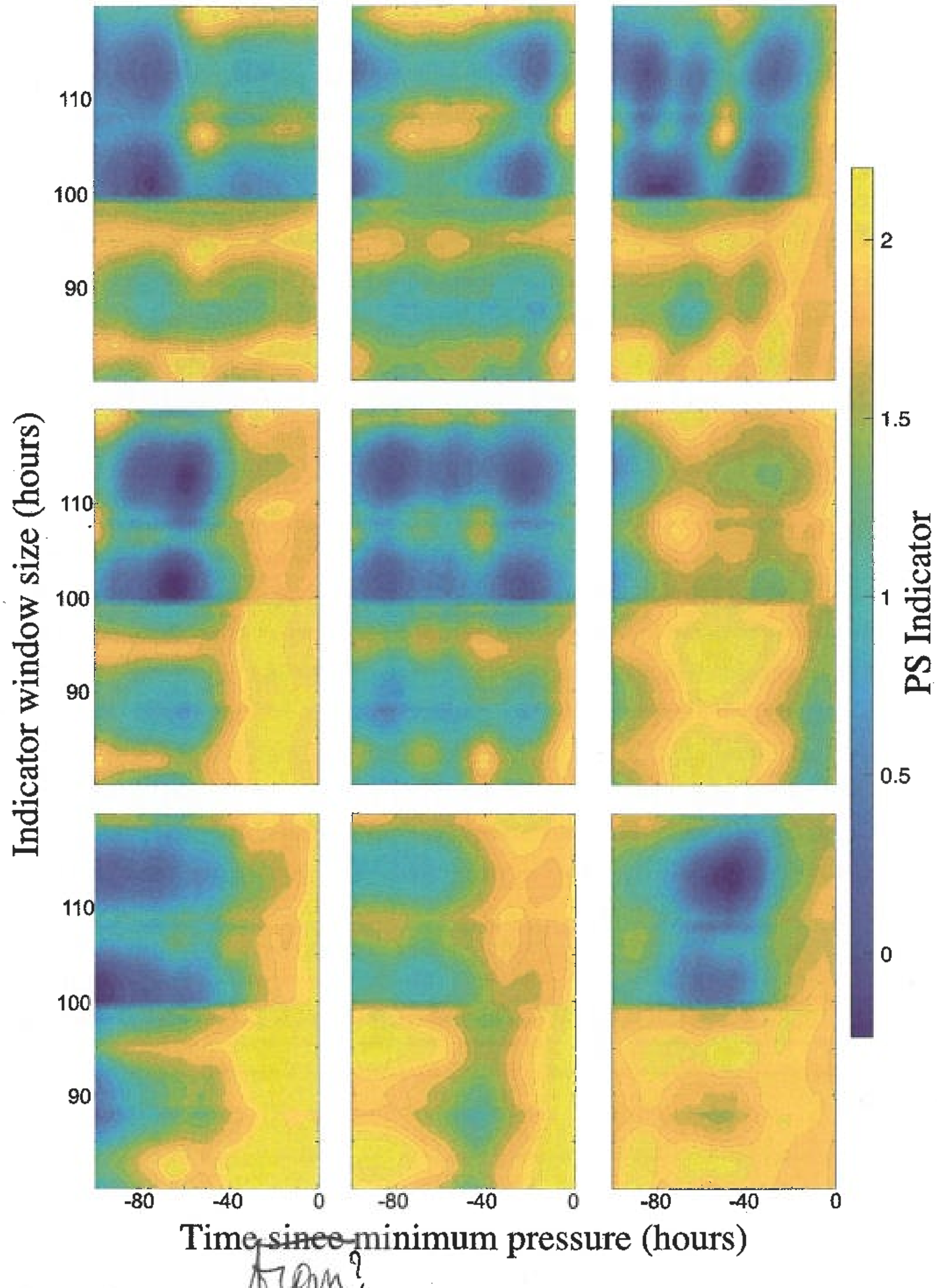


Fig. 4.13 Contour plots showing the sensitivity of the PS indicator to the window size. Reading left to right the plots show the results for: Andrew (Florida); Andrew (Louisiana); Katrina; Wilma; Gustav; Matthew; Harvey; Irma; Nate. In each case, for each window size, the PS indicator is calculated for the sea-level pressure time series given at each station in the landfall region and the mean taken over all of these stations.

sparse observations, particularly the observations of maximum wind speed V_m given by

sampled

$$V_m = \left(\frac{B(p_n - p_c)}{\rho e} \right)^{1/2} \tag{4.7}$$

and occurring at a radius $R_w = A^{1/B}$.

Here we present a similar model modified so that the pressure is modelled at a fixed point in space as a function of time, rather than a profile modelled at a fixed time as a function of the distance from the hurricane centre. We are therefore able to model the effect on sea level pressure (at a weather station, for example) of an approaching hurricane. We use the values $A = 40$ and $B = 1$ which are obtained by fitting to Hurricane Katrina (2005) at peak intensity. We consider a fixed point at distance $d(t)$ from the hurricane centre at time t. In equation 4.6, we replace r by $d(t)$ to introduce a time dependence. We assume the hurricane moves in a straight line towards the fixed point with a speed of $v(t)$kmph, in which case we have $d(t) = \int_0^t v(s)ds$. For this model, a constant velocity will suffice: 18kmph is obtained by taking a mean of the nine tropical cyclones studied in section 4.4 using the HURDAT2 data (see section 4.1.1). We in fact use a velocity ~~chosen~~ uniformly from the range $[16, 20]$ so that not all implementations of the model are identical. The ambient pressure, p_n, is given by

use tilde Km/h

$$p_n(t) = 1016 + 5\eta_1 + 1.6 \sin\left(\frac{2\pi t}{12} + K \right), \tag{4.8}$$

good writing

and

where K is a random offset chosen uniformly in the range $[0, 12]$, the sine wave models the 12-hourly tidal oscillations. The central pressure is given by $p_c = 950 + 10\eta_2$. Values η_1 and η_2 are sampled from a standard normal distribution. The reason for the inclusion of η_1, η_2 and the random offset K is that we may wish to run the model multiple times and take the mean of statistics over those simulations to obtain an ensemble estimate. For this purpose, we would not want all of the sine wave components to be aligned, nor for all the models to have exactly the same initial parameters.

memory (using EWS indicators),

There is also a need to model the increasing ~~PS indicator value~~ shown to exist in the pressure signal. We therefore add a noise signal to the output in which the power spectrum scaling exponent, α, increases over time. The maximum value of α is greater for points closer to the cyclone track, and the time at which the maximum value is reached is the time at which the closest approach occurs, which we call $T_{\min}$. We have used values such that $\alpha = 0.4$ for $t < t_{\min} - 50$ (more than fifty hours before the minimum approach distance is reached). In the 50-hour window ~~before the minimum approach distance,~~ the value of α increases linearly from the background value of 0.4 to a higher value α_0. α_0 takes the value 2 in cases where the hurricane path passes directly over the grid-point in question (an approach distance of zero) and decreases linearly with increasing distance from the cyclone track, to a minimum value of 0.4 for grid-points more than 200km distant.

The part of the noise signal with increasing α value is made by concatenating 10 sub-series, each of length 1000 and with a higher α value than the last (covering the range $[0.4, \alpha_0]$), then sampling with an interval size chosen to give a series of the desired length. All of the noise signals in the model are generated by the ~~coloured noise generation~~ method detailed by Kasdin [1995].

We therefore have the model:

$$p(d(t)) = p_c + (p_n(t) - p_c)\exp\left(\frac{-A}{d(t)^B}\right) + \eta_t, \tag{4.9}$$

where $d(t)$ is distance to the centre of the hurricane at time t, p_c is the central pressure, $p_n(t)$ is the ambient pressure at time t given by equation 4.8, $A = 40$ and $B = 1$ are fitted parameters, and η_t is a noise term with changing power spectrum scaling exponent as described above. For every point on a spatial grid we are able to calculate the distance $d(t)$ using simple geometry (we model the hurricane as moving in a straight line for simplicity), generate the noise series η and then calculate the pressure at that point using equation 4.9.

ignore this

We model sea-level pressure at a single point in space which is approached by a hurricane passing by at a distance of 20km, the model uses a time-step of 0.05 hours and 100 implementations are evaluated. We then calculate the ACF1 and PS indicators in a 100-hour sliding window in order to compare them with the results in figure 4.4. The result is shown in figure 4.15. We see that the ACF1 and PS indicators behave similarly to the result from the real sea-level pressure data, and the pressure series itself also looks reasonable.

The model is then evaluated with a time-step of 0.05 hours at 100 points in a ten-by-ten grid with 40km spatial separation between points (giving a square of side 360km). The hurricane is modelled as travelling from 'south' to 'north' from 400 hours before reaching the bottom of the grid up to the point where it reaches the top of grid. The PS indicator of the pressure signal for each grid-point is calculated in a 100-hour sliding window (of 20,000 points). We then calculate the PS indicator slope, evaluated using the Mann-Kendall coefficient in the exact same way as shown in the previous section (figure 4.14), in the 30-hour window before the hurricane reaches the bottom of the grid. Figure 4.16 shows the mean over ten such evaluations for ten separate runs of the model. We see that the pattern is consistent with our expectations and also with the general pattern of the analysis of real data in figure 4.14. This confirms our understanding of the hurricane statistics

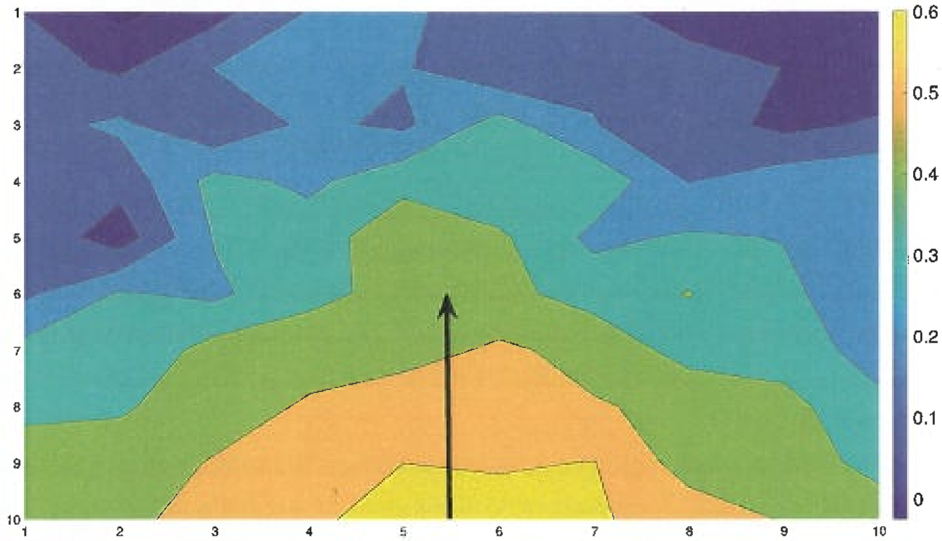


Fig. 4.16 The mean of the PS ~~indiactor~~ indicator slope over ten realisations of the model. The slope is evaluated for the PS indiactor of the pressure signal at each of 100 grid points with 20km spacing, in a 30-hour window before the modelled hurricane reaches the bottom of the image. The ~~forward~~ motion of the hurricane is shown by the black arrow.

4.6 Discussion

We have applied the tipping point indicators introduced in chapters 2 and 3 to the meteorological variables measured in the vicinity on an approaching tropical cyclone. When applied to the sea-level pressure data, the ACF1 and DFA indicators do not appear to provide an EWS. However, the PS indicator clearly does begin to increase around 48 hours before the lowest pressure, about the same point that the decreasing trend in pressure becomes visible, suggesting that it can provide a detectable EWS in this context. The methods used to incorporate wind speed data into the analysis besides sea-level pressure have not appeared to yield any improvement, this is possibly due to the two variables being closely correlated.

We have also attempted to increase the robustness of ~~the~~ results provided by the PS indicator by including data from many points in a close geographical region. This has not produced a reliable early warning signal but ~~has been successful insofar as~~ it has allowed us to visualise an approaching storm by plotting the strength of the increasing trend in the PS indicator over the region. We therefore have confirmed that in this specific system the PS indicator increases consistently prior to and during the tipping event, which is not a bifurcational tipping, whilst the DFA indicator shows no change (see section 4.2). This provides a counter example to claims that the DFA and PS scaling exponents measure essentially the same thing and are not both necessary [Heneghan and McDarby, 2000], possibly a better counter example than the contrived noise series presented in figure 2.3 (section 2.2).

We note that the tropical cyclone example was chosen because it appears to exhibit tipping behaviour whilst not being described, according to our current knowledge, by a bifurcating or state-switching dynamical system often associated with tipping point analysis, therefore presenting the opportunity to apply known methods in a novel system. ~~And also because data is readily available.~~ It was not ~~expected that the tipping point methods available, including the novel PS indicator, would provide a clear early warning~~ signal suitable for ~~prediction. However,~~ In ~~in~~ future work it will be interesting to apply the PS indicator in applications where it may be expected to ~~yield an early warning signal, in which case it can properly~~ be compared to the existing ACF1 and DFA indicators. In particular, the methods applied in section 4.4 to sea-level pressure data over a geographic area could usefully be applied to the greening and desertification of the Sahara, similarly to the methods presented by Bathiany et al. [2013] (discussed in section 3.6). At the time of writing, model data with sufficient temporal resolution are not available.

Another interesting system worth studying with these methods is the formation, rather than the approach, of a tropical cyclone (cyclogenesis) since this is closer to the idea of a complex system undergoing a critical transition [Scheffer et al., 2009] and therefore a better candidate for tipping point analysis. However, real-life data, at a resolution suitable for tipping-point